

In order to correctly model the different possible states of the system, it will be necessary to cover a large part of the accessible phase space, so either trajectories must be very long or we must use many initial conditions. There are many ways to solve initial value problems such as (2.1) combined with an initial condition $\mathbf{z}(0) = \boldsymbol{\zeta}$. The methods introduced here all rely on the idea of a discretization with a finite stepsize h , and an iterative procedure that computes, starting from $\mathbf{z}_0 = \boldsymbol{\zeta}$, a sequence $\mathbf{z}_1, \mathbf{z}_2, \dots$, where $\mathbf{z}_n \approx \mathbf{z}(nh)$. The simplest scheme is certainly Euler's method which advances the solution from timestep to timestep by the formula¹

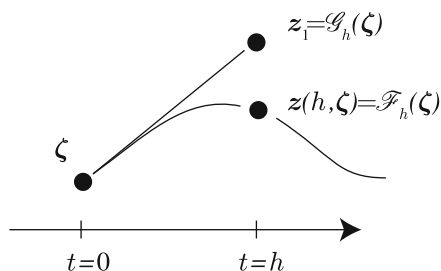
$$\mathbf{z}_{n+1} = \mathbf{z}_n + hf(\mathbf{z}_n).$$

The method is based on the observation that $\mathbf{z}(t+h) \approx \mathbf{z}(t) + h\dot{\mathbf{z}}(t)$, i.e. the beginning of a Taylor series expansion in powers of h , and the further observation that the solution satisfies the differential equation, hence $\dot{\mathbf{z}}(t)$ may be replaced by $\mathbf{f}(\mathbf{z}(t))$.

In order to be able to easily compare the properties of different methods in a unified way, we focus in this chapter primarily on a particular class of schemes, generalized *one-step methods*. Suppose that the system under study has a well defined flow map $\mathcal{F}_t$ defined on the phase space (which is assumed to exclude any singular points of the potential energy function). The solution of the initial value problem, $\dot{\mathbf{z}} = \mathbf{f}(\mathbf{z})$, $\mathbf{z}(0) = \boldsymbol{\zeta}$, may be written $\mathbf{z}(t, \boldsymbol{\zeta})$ (with $\mathbf{z}(0, \boldsymbol{\zeta}) = \boldsymbol{\zeta}$), and the flow-map $\mathcal{F}_t$ satisfies $\mathcal{F}_t(\boldsymbol{\zeta}) = \mathbf{z}(t, \boldsymbol{\zeta})$. A *one-step method*, starting from a given point, approximates a point on the solution trajectory at a given time h units later. Such a method defines a map $\mathcal{G}_h$ of the phase space as illustrated in Fig. 2.1.

We assume here a basic understanding of ordinary differential equations; some good references for review of this topic are [16, 51, 177, 362]. For basic concepts in the numerical analysis of ordinary differential equations the definitive reference is [167].

Fig. 2.1 A step with the flow map approximation $\mathcal{G}_h$ is illustrated in comparison to the corresponding step along the solution curve defined by the flow map $\mathcal{F}_h$



¹Subscripts were used previously to indicate the components of vectors and here they are used to indicate the indexing of timesteps. Although in theory this could lead to some confusion, it normally does not in practice, since we index components in descriptions of details of models and we discuss timesteps in context of defining numerical methods for general classes of systems. Moreover, we use boldface for vectors, so a subscript on a boldface vector indicates a timestep index. When we wish to refer to both the timestep and the component index, we may write $z_{n,i}$ to denote the i th component at timestep n .

$$\mathcal{G}_h(\mathbf{z}) = \mathbf{z} + h\mathbf{f}(\mathbf{z}),$$

and we have

$$\mathbf{z}_{n+1} = \mathcal{G}_h(\mathbf{z}_n), \quad n = 0, 1, 2, \dots \quad (2.2)$$

with $\mathbf{z}_0 = \boldsymbol{\xi}$. The points $\{\mathbf{z}_n\}$ are intended to be approximations to the values of the solution. The obvious question is this: how good an approximation is $\mathbf{z}_n$ of $\mathbf{z}(nh)$?

Let the approximate solution vectors at successive timesteps be $\mathbf{z}_0, \mathbf{z}_1, \dots, \mathbf{z}_v$ where $vh = \tau$. We assume that τ , the length of the time interval, is fixed, and v is an integer parameter representing the total number of timesteps. In order to improve the quality of the approximation, the parameter v may be increased, as the stepsize is proportionately decreased. The error at step n is defined by $e_n = \|\mathbf{z}_n - \mathbf{z}(t_n)\|$, where $t_n = nh$; it clearly depends on h . With these definitions one can prove the following result, which is typical of the sorts of results that are available for numerical methods for initial value problems:

Theorem 2.1 *Let $\mathcal{D}$ be a bounded, open region in $\mathbb{R}^m$ such that $\mathbf{f} : \mathcal{D} \rightarrow \mathbb{R}^m$ is continuously differentiable. Let $\boldsymbol{\xi}$ be an interior point of $\mathcal{D}$ and suppose the initial value problem (2.1) has a unique solution that remains in $\mathcal{D}$ for $t \in [0, \tau]$. Then there exists a constant $C(\tau) > 0$ such that for sufficiently large $v \in \mathbb{N}$ the numerical solution $\mathbf{z}_n$ remains in $\mathcal{D}$ for $n = 0, 1, \dots, v$, where $hv = \tau$, and, moreover, the maximum global error in Euler's method satisfies*

$$\bar{e} := \max_{0 \leq n \leq v} e_n \leq C(\tau)h.$$

In short, the error in the approximation obtained on $[0, \tau]$ is reduced in direct proportion to the number of steps taken to cover this interval. Another way to say this is that $\bar{e} \propto h$, or, using the order notation, $\bar{e} = \mathcal{O}(h)$. Because the global error is of order h^r , where $r = 1$, we say that Euler's method is a *first order* method, or that it *converges with order* $r = 1$.

2.1.1 Example: The Trimer

Let us test Euler's method on the trimer model (formulated as a system in $\mathbb{R}^4$). Fixing initial values ($\mathbf{q}_0 = (0.5, 1.0); \mathbf{p}_0 = (0.1, 0.1)$) we solve the trimer on $[0, 4]$ using the Euler method first with 40 time-steps of length 0.1, next using 80 time-steps of length 0.05 and then with 160 steps of size 0.025, etc. Each calculation results in a different “discrete trajectory.” The solutions are graphed in the xy projection in the left panel of Fig. 2.2 with line segments connecting the points; these piecewise linear paths take on the appearance of smooth curves as the stepsize is reduced.